

A Counterexample to General Convergence of Double Q-Learning with Linear Function Approximation

1 Claim

Double Q-learning is convergent in the tabular case under standard assumptions. This proof shows that the same is not true for arbitrary linear function approximation.

Theorem 1. *There exists a finite discounted Markov decision process, a stationary behavior policy, a bounded linear feature map, and a Robbins–Monro stepsize sequence such that symmetric Double Q-learning with linear function approximation is unbounded almost surely from any strictly positive initialization. This holds even though the optimal action-value function is identically zero and is exactly representable by the feature map.*

The example has one state, two actions, zero rewards, and one scalar feature.

2 The MDP, features, and behavior policy

Consider the discounted MDP with one state s , two actions a_1, a_2 , deterministic self-transition, zero rewards, and discount factor $\gamma = 0.9$. Thus $P(s \mid s, a) = 1$ and $r(s, a) = 0$ for both actions $a \in \{a_1, a_2\}$. Since all rewards are zero,

$$Q^*(s, a_1) = Q^*(s, a_2) = 0. \quad (1)$$

Use one-dimensional linear features $\phi(s, a_1) = 1$ and $\phi(s, a_2) = 2$. The two Double-Q estimators are parameterized by scalars θ_A, θ_B :

$$Q_A(s, a) = \theta_A \phi(s, a), \quad Q_B(s, a) = \theta_B \phi(s, a). \quad (2)$$

The optimal function $Q^* \equiv 0$ is exactly representable by the parameter value $\theta_A = \theta_B = 0$.

Let the stationary behavior policy be $\mu(a_1 \mid s) = 0.9$ and $\mu(a_2 \mid s) = 0.1$. At time n , sample an action from μ , and write $X_{n+1} = \phi(s, A_{n+1}) \in \{1, 2\}$. Hence

$$\mathbb{P}(X_{n+1} = 1) = 0.9, \quad \mathbb{P}(X_{n+1} = 2) = 0.1. \quad (3)$$

Independently of the sampled action and of the past, choose estimator A or estimator B for update with probability $1/2$ each. Use the global stepsize sequence

$$\alpha_n = \frac{a}{n+1}, \quad 0 < a \leq \frac{1}{16}. \quad (4)$$

Then $\sum_{n=0}^{\infty} \alpha_n = \infty$ and $\sum_{n=0}^{\infty} \alpha_n^2 < \infty$. Initialize with arbitrary strictly positive parameters: $\theta_{A,0} > 0$ and $\theta_{B,0} > 0$.

3 The Double Q-learning updates

We use the standard symmetric Double Q-learning update. If estimator A is selected at time n , then

$$\theta_{A,n+1} = \theta_{A,n} + \alpha_n \phi(s, A_{n+1}) [\gamma Q_B(s, \arg \max_{a'} Q_A(s, a')) - Q_A(s, A_{n+1})], \quad (5)$$

while θ_B is unchanged. If estimator B is selected, the roles are reversed:

$$\theta_{B,n+1} = \theta_{B,n} + \alpha_n \phi(s, A_{n+1}) [\gamma Q_A(s, \arg \max_{a'} Q_B(s, a')) - Q_B(s, A_{n+1})], \quad (6)$$

while θ_A is unchanged. Because the state never changes and the reward is zero, these are the full updates.

4 Positivity is invariant

We first show that the positive quadrant is invariant. This will imply that the greedy action is fixed throughout the entire trajectory.

Suppose $\theta_A > 0$. Then $Q_A(s, a_2) = 2\theta_A > \theta_A = Q_A(s, a_1)$, so the greedy action under Q_A is uniquely a_2 . Similarly, if $\theta_B > 0$, the greedy action under Q_B is uniquely a_2 . Thus no tie-breaking convention is needed as long as the parameters remain strictly positive.

Assume $\theta_A, \theta_B > 0$, and let $X \in \{1, 2\}$ be the sampled feature. If estimator A is updated, then the target action is a_2 , whose feature is 2. Therefore

$$\theta_A^+ = \theta_A + \alpha X (\gamma \cdot 2\theta_B - X\theta_A) = (1 - \alpha X^2)\theta_A + 1.8\alpha X \theta_B. \quad (7)$$

Because $X \leq 2$ and $\alpha_n \leq 1/16$, we have $1 - \alpha_n X^2 \geq 1 - 4\alpha_n \geq 3/4 > 0$. Thus $\theta_A^+ > 0$. The same argument applies when estimator B is updated:

$$\theta_B^+ = (1 - \alpha X^2)\theta_B + 1.8\alpha X \theta_A > 0. \quad (8)$$

Therefore, since $\theta_{A,0} > 0$ and $\theta_{B,0} > 0$, we have $\theta_{A,n} > 0$ and $\theta_{B,n} > 0$ for every n . Consequently, the greedy action is always a_2 for both estimators. Along this trajectory, the nonlinear maximization in the Double-Q target reduces to a fixed linear target.

5 Dynamics of the parameter sum

Define $S_n = \theta_{A,n} + \theta_{B,n}$. By positivity invariance, $S_n > 0$ for all n . We will prove that $S_n \rightarrow \infty$ almost surely. This implies that the Double-Q parameter vector is unbounded and therefore cannot converge to $(0, 0)$, which represents Q^* .

Let $r_n = \theta_{A,n}/S_n \in (0, 1)$. If estimator A is updated at time n , then

$$S_{n+1} = S_n + \alpha_n X_{n+1} (1.8\theta_{B,n} - X_{n+1}\theta_{A,n}). \quad (9)$$

Dividing by S_n , this becomes $S_{n+1} = S_n(1 + \alpha_n Y_{n+1})$, where, in the case of an A -update,

$$Y_{n+1} = X_{n+1} (1.8(1 - r_n) - X_{n+1} r_n). \quad (10)$$

If estimator B is updated, then similarly $S_{n+1} = S_n(1 + \alpha_n Y_{n+1})$, where

$$Y_{n+1} = X_{n+1} (1.8r_n - X_{n+1}(1 - r_n)). \quad (11)$$

For both possible updates, because $r_n \in (0, 1)$ and $X_{n+1} \in \{1, 2\}$, we have $-4 \leq Y_{n+1} \leq 3.6$. Therefore $|\alpha_n Y_{n+1}| \leq 1/4$ and $1 + \alpha_n Y_{n+1} \geq 3/4 > 0$. Thus the logarithm of S_n is well defined for every n .

6 Positive drift of the logarithm

Let $\mathcal{F}_n$ be the history up to time n , including $\theta_{A,n}$ and $\theta_{B,n}$, but not the fresh action sample or estimator choice at time n . Conditional on $\mathcal{F}_n$, the ratio r_n is fixed.

First average over the independent choice of which estimator is updated. For a fixed sampled feature X ,

$$\begin{aligned}\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n, X_{n+1} = X] &= \frac{1}{2} X (1.8(1 - r_n) - X r_n) + \frac{1}{2} X (1.8 r_n - X(1 - r_n)) \\ &= \frac{1}{2} X (1.8 - X).\end{aligned}\tag{12}$$

Now average over the behavior policy:

$$\begin{aligned}\mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] &= 0.9 \cdot \frac{1}{2} \cdot 1 \cdot (1.8 - 1) + 0.1 \cdot \frac{1}{2} \cdot 2 \cdot (1.8 - 2) \\ &= 0.9 \cdot 0.4 + 0.1 \cdot (-0.2) \\ &= 0.36 - 0.02 = 0.34.\end{aligned}\tag{13}$$

Thus Y_{n+1} has a strictly positive conditional mean, uniformly over the current parameters.

Define $L_n = \log S_n$. Since $S_{n+1} = S_n(1 + \alpha_n Y_{n+1})$, we have $L_{n+1} - L_n = \log(1 + \alpha_n Y_{n+1})$. For $|t| \leq 1/4$, the elementary inequality $\log(1 + t) \geq t - t^2$ holds. Applying this with $t = \alpha_n Y_{n+1}$, and using $|Y_{n+1}| \leq 4$, gives

$$\mathbb{E}[L_{n+1} - L_n \mid \mathcal{F}_n] \geq \alpha_n \mathbb{E}[Y_{n+1} \mid \mathcal{F}_n] - \alpha_n^2 \mathbb{E}[Y_{n+1}^2 \mid \mathcal{F}_n] \geq 0.34 \alpha_n - 16 \alpha_n^2.\tag{14}$$

Because $\sum_{n=0}^{\infty} \alpha_n = \infty$ and $\sum_{n=0}^{\infty} \alpha_n^2 < \infty$, we have $\sum_{n=0}^{N-1} (0.34 \alpha_n - 16 \alpha_n^2) \rightarrow +\infty$. So the predictable drift of L_n has a deterministic lower bound whose partial sums diverge to $+\infty$.

7 Martingale fluctuations are negligible

Let $\Delta L_{n+1} = L_{n+1} - L_n = \log(1 + \alpha_n Y_{n+1})$, and define the martingale difference

$$M_{n+1} = \Delta L_{n+1} - \mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n].\tag{15}$$

For $|t| \leq 1/4$, there is a universal constant K such that $|\log(1 + t)| \leq K|t|$. For example, one may take $K = 2$. Therefore $|\Delta L_{n+1}| \leq 2\alpha_n |Y_{n+1}| \leq 8\alpha_n$. Consequently,

$$|\mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n]| \leq \mathbb{E}[|\Delta L_{n+1}| \mid \mathcal{F}_n] \leq 8\alpha_n.\tag{16}$$

Hence $|M_{n+1}| \leq |\Delta L_{n+1}| + |\mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n]| \leq 16\alpha_n$. It follows that $\mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] \leq 256 \alpha_n^2$. Since $\sum_n \alpha_n^2 < \infty$,

$$\sum_{n=0}^{\infty} \mathbb{E}[M_{n+1}^2 \mid \mathcal{F}_n] < \infty.\tag{17}$$

By the martingale convergence theorem for square-integrable martingales with summable conditional variances, the martingale partial sums $\sum_{n=0}^{N-1} M_{n+1}$ converge almost surely to a finite random limit.

8 Almost-sure divergence

Using the decomposition

$$L_N = L_0 + \sum_{n=0}^{N-1} \mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n] + \sum_{n=0}^{N-1} M_{n+1}, \quad (18)$$

and the lower bound from Section 6,

$$\sum_{n=0}^{N-1} \mathbb{E}[\Delta L_{n+1} \mid \mathcal{F}_n] \geq \sum_{n=0}^{N-1} (0.34 \alpha_n - 16 \alpha_n^2) \rightarrow +\infty. \quad (19)$$

The martingale term converges almost surely to a finite limit. Therefore $L_N \rightarrow +\infty$ almost surely. Since $L_N = \log S_N$, this implies $S_N \rightarrow \infty$ almost surely. Since both parameters remain positive,

$$\max\{\theta_{A,N}, \theta_{B,N}\} \geq \frac{S_N}{2} \rightarrow \infty. \quad (20)$$

Thus the parameter vector is unbounded almost surely. In particular, the Double-Q iterates cannot converge to the representable optimum $(\theta_A, \theta_B) = (0, 0)$. This proves the theorem. $\square$

9 Relation to the expected linear dynamics

This section is not needed for the almost-sure proof, but it explains the source of the instability. In the positive quadrant, the greedy action is always a_2 . Therefore the expected update is linear. Let $C = \mathbb{E}[X^2]$ and $D = \gamma \mathbb{E}[2X]$. In this example, $C = 0.9 \cdot 1^2 + 0.1 \cdot 2^2 = 1.3$ and $D = 0.9(0.9 \cdot 2 \cdot 1 + 0.1 \cdot 2 \cdot 2) = 1.98$. The expected Double-Q update has the form

$$\mathbb{E}[\Delta \theta_A] = \frac{\alpha}{2}(D \theta_B - C \theta_A), \quad \mathbb{E}[\Delta \theta_B] = \frac{\alpha}{2}(D \theta_A - C \theta_B). \quad (21)$$

Equivalently,

$$\mathbb{E} \begin{bmatrix} \Delta \theta_A \\ \Delta \theta_B \end{bmatrix} = \frac{\alpha}{2} \begin{bmatrix} -C & D \\ D & -C \end{bmatrix} \begin{bmatrix} \theta_A \\ \theta_B \end{bmatrix}. \quad (22)$$

Introduce the average and difference coordinates $u = (\theta_A + \theta_B)/2$ and $v = (\theta_A - \theta_B)/2$. The corresponding mean ODE decouples as

$$\dot{u} = \frac{1}{2}(D - C)u, \quad \dot{v} = -\frac{1}{2}(C + D)v. \quad (23)$$

Since $D - C = 1.98 - 1.3 = 0.68 > 0$, we obtain $\dot{u} = 0.34u$. Thus the average mode is unstable. The almost-sure argument above turns this mean-dynamics instability into a genuine stochastic nonconvergence result.

10 What the counterexample shows and does not show

This counterexample shows that there cannot be a general convergence theorem for Double Q-learning with arbitrary linear function approximation under off-policy sampling and bootstrapping.

It does not show that Double Q-learning always diverges. Double Q-learning is convergent in tabular settings under the usual assumptions, and it may converge in restricted linear settings with additional structure. The point is that Double Q-learning does not, by itself, remove the instability caused by the combination of bootstrapping, off-policy sampling, and function approximation.

The example is deliberately minimal: one state, two actions, zero rewards, one scalar feature, and a fixed behavior policy. Despite this simplicity, the Double-Q parameter norm diverges almost surely from every strictly positive initialization.